

Lab 1: Entity-Relationship Diagram (ERD), Define and Design

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1 Objectives

The purpose of this exercise is to practice how to define and design database structures. We do that through, successfully, identifying entities from a provided scenario. After we have defined the entities and roles, we proceed with designing the conceptual model first and based on it even the logical model.

2 Reading

To be able to actively participate in this lab the student is expected to have read the lectures. Additional reading from provided resources is beneficial but not crucial to the completion of this lab.

3 Setup

There are no special requirements when it comes to tools used. The most common practice is to draw your ER diagram on paper. However, you will have to hand in your diagram in a digital format via Canvas. Here, we recommend using "draw.io" (available online, will be used on the exam), but you can of course use any other website/tool to make your drawings digital.

4 Presentation

The presentation for Lab 1 is done orally. **All group members** present to the lab assistant which then determines if the demonstrated knowledge is sufficient for a pass. If not, the students can present at a later occasion, same or other day. The lab is graded individually, meaning that one team member can pass while the other fails.

5 Instructions

You must work in a **group of 2 students**, together, but with equal work distribution. Your team will be the same for all the remaining laboratory assignments as well. You must be with a partner that is in the same lab session as you (easier for everybody). Based on the scenario (system specification) your task is to **design an Entity Relationship Diagram (ERD)** which will later lead to a database. As it usually is in practice, no system specification you come across will be complete and perfect. It is your task to identify gaps in the system and try to complete them with preferred solutions. It is just important that, during your presentation, you motivate why you chose to use that specific solution.

To make things easier for you the system in question will be a university database system so feel free to append own ideas and experiences on the given scenario. The hand-in will consist of your conceptual model as a draw.io, PNG or PDF file. Important things you should consider while drawing your models:

- Entity types
- Relationship types
- Attributes
- Cardinality (please use the notation min:max defined in the lecture)

Please respect the notation used in the lecture (see Figure 1). If needed, you can also use the concept of inheritance (see Figure 2).

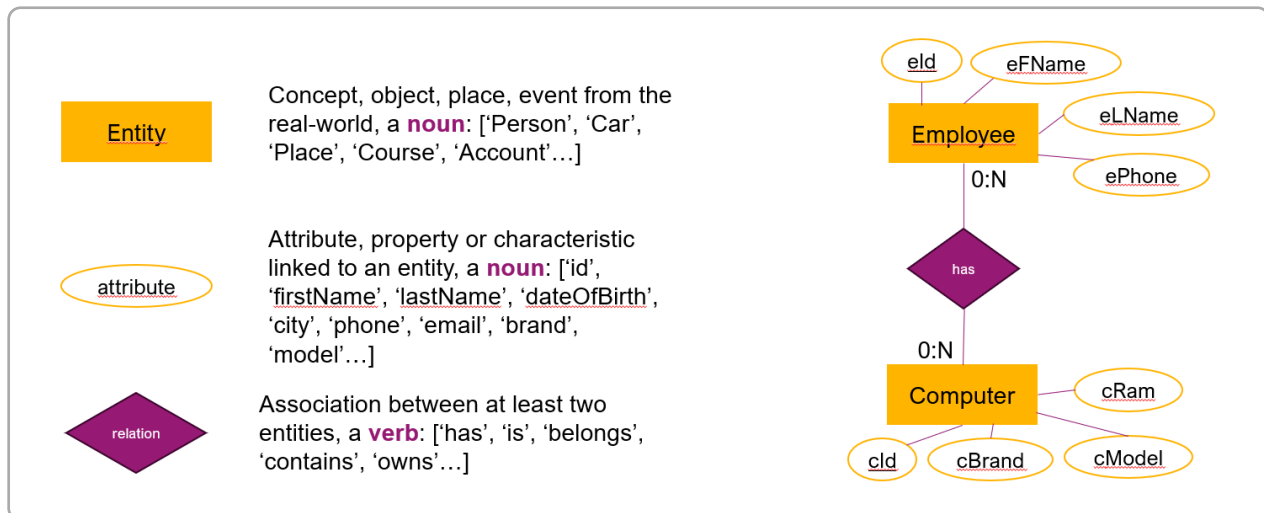


Figure 1: Entity-Relationship Diagram (ERD) components.

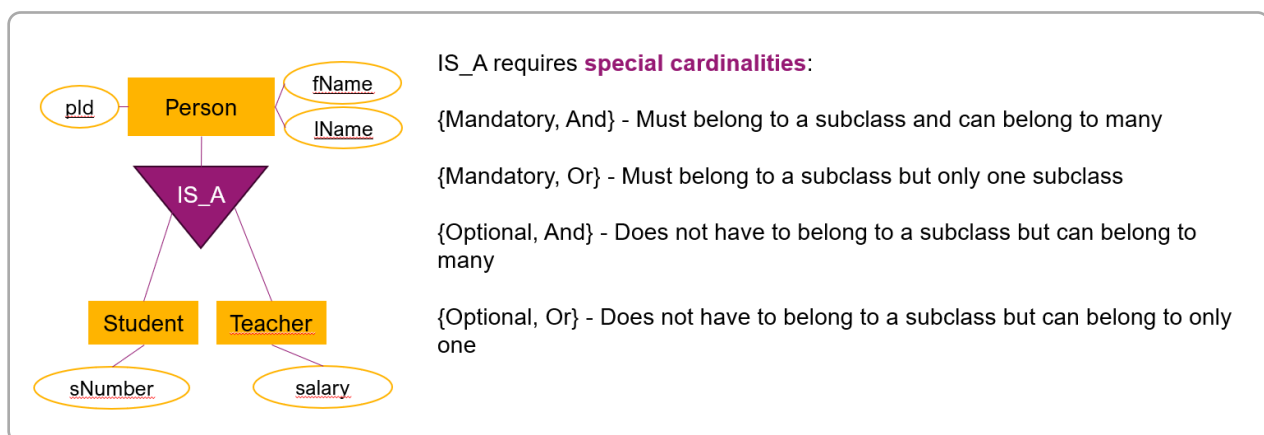


Figure 2: The concept of inheritance.

6 Scenario

A university has several schools. It has to have at least one school but can have many different schools under its umbrella. These schools can be such as "School of Education and Communication", "School of Engineering" etc. Each school has basic information about it, such as address, dean, number of students, departments and so on. The departments offer courses and programmes. A department has a name, e.g. "Computer Science and Informatics", "Materials Engineering", "Building and Lighting" etc., a code such as "CSI0054" and a number of employees (staff). The school offers educational programmes to students, and these programmes belong to a specific department. One department can have 0 or many programmes. Some programmes belonging to, for example, the "Computer Science and Informatics" department are "Software Engineering and Mobile Platforms", "Embedded systems", "AI Engineering" etc. Each programme has a programme code, e.g. TGMM7 and a programme short name e.g. DMP. These are differentiated by their start year, which is most often autumn, but can also be spring. Hence, the most common notation is A19 or S20 where the numbers 19 and 20 show the year the students started. There are different types to programmes, they can be "Master of Science", "Bachelor of Science", "Bachelor of Arts", "Candidate", etc. A programme must consist of a number of courses to classify as a programme. It has to have a minimum of 5 courses but can have many courses. A single course can be given to multiple programmes, but really does not have to belong to any programme for that matter. Also, a course can be mandatory or elective. To be able to identify a course the university uses a course name, e.g. "Databases" but also a unique course code "TDRK18". Apart from that the course offers a certain amount of ECTS (credits) upon completion.

Each school has employees, that ultimately belong to departments. One individual can work at one or many departments. All employees at the school have identified roles, such as "Dean", "Head of Education", "Head of Department", "Programme Manager", "Lecturer" and so on. One employee must have one role but can also have many roles. Employees usually have a personal number, employee number, address, email, work phone, private phone, title (which is different than role, can be Dr., MSc., BSc., Prof.), salary, office number, date of employment. Each department has one employee serving as a Head of Department, programme has one employee serving as a programme manager and courses have one employee as course manager. A course has one or many lecturers. A course must have a course manager, must have an examiner and a lecturer. One lecturer can be giving lectures in different courses. Ultimately, one employee can be a head of department, programme manager, course manager and also a lecturer at one course at the same time. One employee can be a course manager for many different courses. Extra salary bonus and this is different for different roles. Not all roles have this bonus.

A course can have one or more examinations, such as exam, assignment, labs. A specific course does not have to necessarily have examinations of any kind. A specific examination type is shared between courses, for example many courses have exams, labs etc. Each examination part carries a specific amount of ECTS (hp), e.g. Databases has Exam-1 for 2 ECTS, Exam-2 for 2 ECTS, Labs for 2 ECTS, which is a total of 6 ECTS. Each examination must be graded using the 5/4/3/Fail. When a student starts at a university, he/she applies for a specific programme and is later associated (belongs to) that specific programme. A student has a first name, last name, personal number, phone, address, email, username and password. A student reads all courses that belong to their programme. When a student reads a course, and does an examination, a special date is registered for that examination with the examination type and grade. When a student finishes all examinations, a final course grade is entered with a final date.
